

Weekly Report

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Status Since early December

- No major expansion of codebase since early December.
- Focused on stabilizing and tightening the **conceptual foundation**.
- Clarified the exact correspondence between:
 - algebraic K-theory notions, and
 - concrete failure modes in multi-field ER.
- Result: clearer problem decomposition and stronger methodological claims.

Key Conceptual Refinement

Shift in framing:

Before

K-theory provides an abstract language for ER consistency.

Now

ER errors are **structural violations**:

- non-commuting diagrams,
- broken exact sequences,
- invalid pushouts.

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This reframes ER from threshold tuning to **consistency verification**.

Decomposition of the ER Problem

I separated three roles that were previously conflated:

- ① **Pairwise similarity** $\rightarrow$ morphisms between records.
- ② **Multi-field agreement** $\rightarrow$ commutativity / homotopy.
- ③ **Cluster validity** $\rightarrow$ exactness or pushout existence.

This separation clarifies where traditional ER fails and where CEER intervenes.

Negative Insight: Not All Contradictions Should Be Resolved

- Traditional ER tries to reconcile conflicting similarities.
- Structural view:
 - some contradictions indicate **invalid cluster hypotheses**.
- Non-commuting triangles provide a principled reason to **block merges**.

This aligns with observed over-merging in benchmarks like Cora.

Concrete Example: Cora Dataset

Typical failure pattern:

- Title similarity supports $A \sim B$.
- Author similarity supports $B \sim C$.
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Structural interpretation:

$$g \circ f \neq h$$

The triangle does not commute $\Rightarrow$ reject joint clustering.

Concrete Example: Affiliation Strings

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- “MIT CSAIL”
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If $A \simeq B$ and $A \simeq C$, then:

- a pushout must exist,
- otherwise the merge is structurally unjustified.

What Is New Relative to Dec 5

- Sharper distinction between:
 - similarity scores vs. structural validity.
- Explicit acknowledgment of **negative results**:
 - some matches must be rejected, not repaired.
- Clearer path to formalizing homotopy consistency as a model constraint.

Next Technical Step

Planned implementation direction:

- Introduce a **homotopy-consistency penalty** into CEER.
- Penalize non-commuting multi-field triangles.
- Evaluate effect on:
 - over-merging (Cora),
 - inconsistency under noise (NC Voter).

Goal: consistency-regularized ER, not just embedding-based ER.

- ER can be viewed as constructing a homotopy-consistent equivalence structure.
- K-theoretic notions provide **structural rejection criteria**.
- This leads to a principled framework:

Homotopy-Consistent Entity Resolution